

Feedback on C40018 Q1 - Discrete Mathematics - 2025

This year's exam has been of average difficulty with respect to previous years, but had certain parts that a lot of students found really difficult.

Average performance: A^- .

- a) This part asked for some set construction, testing the understanding of the basic set operations; most students got this right.
- b) This part on relations was a variant of an exercise in the notes; most students got this right. Only one part needed a proof, which most got correct; some students mixed up 'complement' and 'inverse', or struggled to construct a counterexample.
- c) This on maximums etc. was a variant of a question that appeared in the coursework. Many did not see that $V_1 = \mathbb{N} \setminus \{0\}$, and $V_0 = \{0\}$; some students did not know that 0 is in $\mathbb{N}$.

Remarkably, many students did not read the question correctly, and treated the relation R (the main object to study in the question) as defined on $W \subseteq \wp \mathbb{N}$, so looked at statements like $V_q R V_p$; R was clearly defined on $\mathbb{N}$, through $p R q \triangleq V_q \subseteq V_p$, so those students did not answer the question correctly.

- d) This part on functions was an exercise from the notes, and most students got this correct.
- e) This part on countability is an exercise in the notes, and was discussed during the lectures. Only few students got this correct.